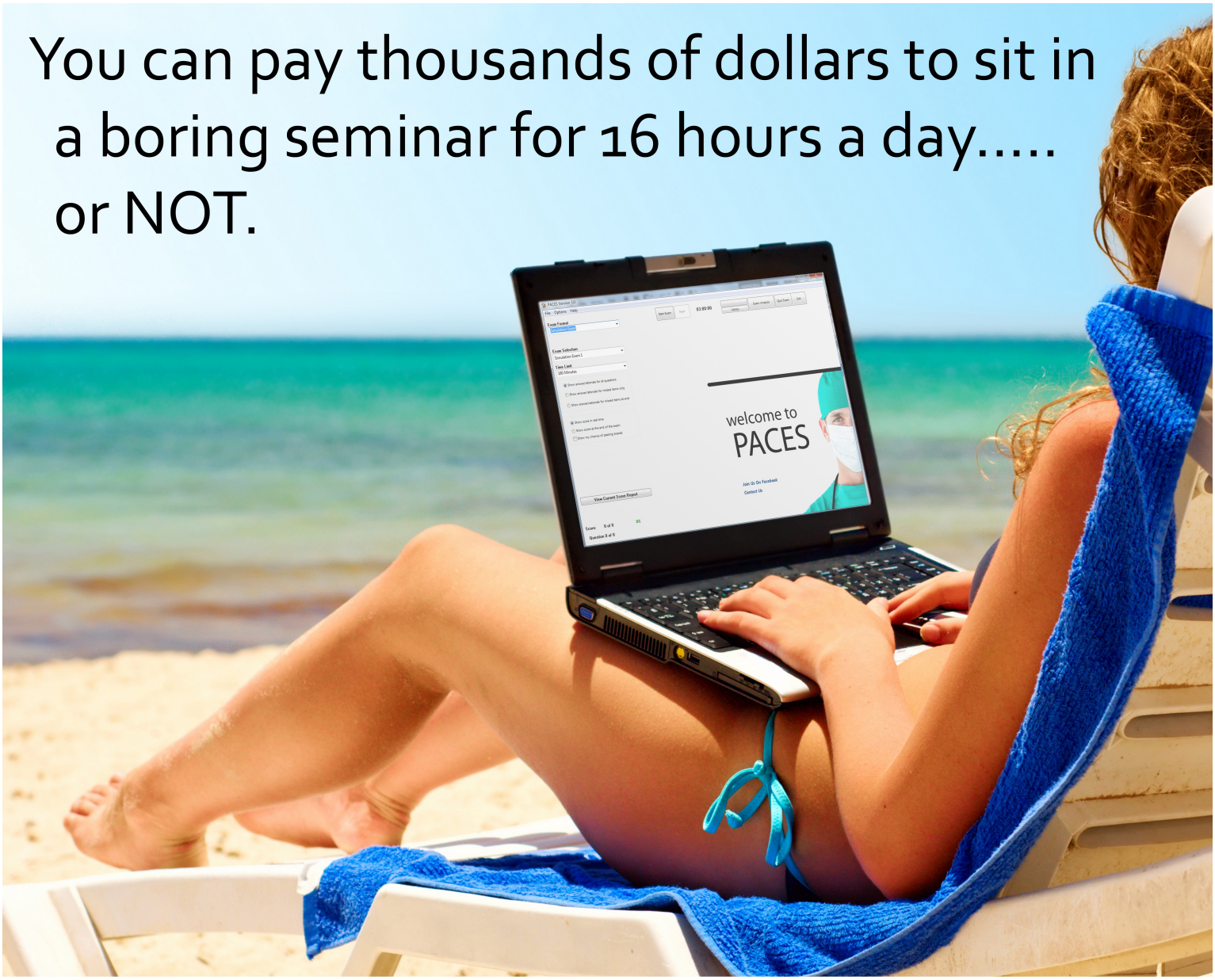
Calculating A-a Gradients



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Calculating A-a Gradients

Although the difference between ventilation and perfusion in the lungs (V/Q mismatch) can be assessed in the laboratory setting using radioactive gases, in actual practice, it is more useful to assess V/Q mismatches by calculating the alveolar-arterial oxygen gradient by subtracting the arterial oxygen (PaO_2) from the 'ideal' alveolar oxygen (PAO_2). The normal A-a gradient in non-smokers is less than 10 mmHg (some sources cite < 15 mmHg). It will also increase about 1 mmHg for each decade of life. A useful estimate of the normal A-a gradient is Age/4 + 4. Thus, the normal A-a gradient for an 80 year-old would be 80/4 + 4 or 20 mmHg.

Depending on the amount of information you are given in the problem, however, there are potentially THREE formulas you must solve to calculate the A-a gradient. This makes the calculations seem complex, but by focusing on only one step at a time and practicing enough problems that you can easily recall the formulas, you can solve these questions reliably. This workbook will take you carefully through each step, show you several examples of each step, demonstrate several complex problems, and then provide you with numerous problems to work on your own.

STEP ONE

$$\text{A-a gradient} = PAO_2 - PaO_2$$

The ideal PAO_2 is what would exist if there were NO ventilation-perfusion defects and the lung had completely normal function. At room air, the normal PAO_2 is about 100-105 mmHg. So, let's imagine the following problems:

What is the A-a gradient of a patient who has a PAO_2 of 105 mmHg and an arterial PO_2 of 95 mmHg?

A-a Gradient = 105 mmHg – 95 mmHg
A-a Gradient = 10 mmHg

What is the A-a gradient of a patient who has a PAO_2 of 99 mmHg and a PaO2 of 71 mmHg?

A-a Gradient = 99 mmHg – 71 mmHg
A-a Gradient = 28 mmHg

What is the A-a gradient of a patient who has a PAO_2 of 159 mmHg and a PaO2 of 101 mmHg?

A-a Gradient = 159 mmHg – 101 mmHg
A-a Gradient = 58 mmHg

What is the A-a gradient of a patient who has a PAO_2 of 143 mmHg and a PaO2 of 131 mmHg?

A-a Gradient = 143 mmHg – 131 mmHg
A-a Gradient = 12 mmHg

That's it! We could only hope that a question this easy would appear on boards. The unfortunate fact, however, is that the PAO_2 is rarely provided to us that easily. If we are not provided the PAO2, then we must calculate it from the information provided. The next page will take the next step in complexity in calculating A-a gradients.

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STEP TWO

If we are presented with a problem in which the PAO_2 is not provided for us, then we must proceed to step two in order to calculate it. The calculation used to arrive at the PAO_2 is shown below:

$$PAO_2 = PIO_2 - \frac{PACO_2}{R}$$

This formula looks much more complicated than the calculation in Step One. So, before we get started, it is helpful to explain the variables used in this formula so we know what values to put in their place when performing the calculation:

PAO_2 This is the alveolar oxygen level and the value we are trying to determine so that we can compare to the arterial oxygen and arrive at the alveolar-arterial oxygen gradient.

PIO_2 This is the inspired oxygen level and is measured in mmHg. If you are provided with the percent oxygen, you must calculate the inspired oxygen in mmHg first. The PIO_2 is potentially the third calculation required in calculating the A-a gradient. Because the units of PIO2 are mmHg and the inspired oxygen content is typically provided in %, you can get an idea of what we'll have to do to arrive at our answer.

$PACO_2$ This is the alveolar carbon dioxide level. The convention is to use the arterial carbon dioxide level in mmHg for this value. In other words, even though the technically most accurate method would use the $PACO_2$ in the formula, we use the $PaCO_2$ because it is much easier to obtain in clinical situations.

R R represents the respiratory exchange ratio which is the ratio of VCO_2/VO_2. Because you inhale more molecules of oxygen than you exhale carbon dioxide, this ratio is normally less than 1. The physiologic value considered to be normal is 0.8. If you are not supplied with a value for R, the use the value 0.8 in your calculations. Because the value for R expresses the difference in oxygen inspired and carbon dioxide exhaled, it is also an expression of metabolism. For example, the use of fat as the primary source of fuel can produce an R closer to 0.7 while the use of carbohydrates as fuel (such as occurs during intense exercise) can produce an R greater than 1.

Notes: Note that some formulas will exhibit another factor in the formula known as F. This is a small calculated factor used as a correctional device when a patient is breathing air. Because this factor is so small, it is traditionally ignored in standard A-a gradient calculations and will be ignored in ours. The point is, if you come across more complex versions of the formula to calculate PAO_2, just ignore them. They may be used in respiratory labs where actual measurements of lung volumes and gas concentrations can be performed. The formula we will be using is the one most relevant to the clinician who has access to simpler values such as arterial blood gases and inspired oxygen concentrations.

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STEP TWO CONT.

Let's examine some example problems:

What is the A-a gradient who has a PaO2 of 80 mmHg and a PaCO2 of 45 mmHg and the PIO_2 is 175 mmHg? (R = 0.8) Express your answer in the nearest whole number.

Examining the problem, we see that the PAO2 isn't provided, so we must calculate it.

$PAO_2 = PIO_2 - (PACO_2/R)$

$PAO_2 = 175 - (45/0.8)$

$PAO_2 = 175 - (56.25)$

$PAO_2 = 118.75$ mmHg

Now, we can insert our calculated PAO_2 into the A-a Gradient formula.

A-a Gradient = $PAO_2 - PaO_2$

A-a Gradient = 118.75 – 80

A-a Gradient = 38.75 mmHg or 39 mmHg

Since we know that the normal value is less than 15 mmHg, this is obviously an elevated A-a gradient.

What is the A-a gradient who has a PaO2 of 124 mmHg and a PaCO2 of 59 mmHg and the PIO_2 is 223 mmHg? (R = 0.8) Express your answer in the nearest whole number.

Examining the problem, we see that the PAO2 isn't provided, so we must calculate it.

$PAO_2 = PIO_2 - (PACO_2/R)$

$PAO_2 = 223 - (59/0.8)$

$PAO_2 = 223 - (73.75)$

$PAO_2 = 149.25$ mmHg

Now, we can insert our calculated PAO_2 into the A-a Gradient formula.

A-a Gradient = $PAO_2 - PaO_2$

A-a Gradient = 149.25 – 124

A-a Gradient = 25.25 mmHg or 25 mmHg

Is this value normal or elevated? If you remember from the introductory paragraph of this workbook, you know that the normal value really depends upon the age of the patient. Let's say the patient is 88 years old. The normal A-a gradient is age/4 + 4 or 88/4 + 4 or 26 mmHg. Although an A-a gradient of 25 mmHg is elevated in a 30 year old patient, it would be considered normal in an 88 year old. Remember this concept, because our practice questions may get a little trickier later on.

We'll do a few more questions on the next page before moving on to Step Three.

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What is the A-a gradient who has a PaO2 of 91 mmHg and a PaCO2 of 44 mmHg and the PIO_2 is 149 mmHg? (R = 0.8) Express your answer in the nearest whole number.

Examining the problem, we see that the PAO2 isn't provided, so we must calculate it.

$PAO_2 = PIO_2 - (PACO_2/R)$

$PAO_2 = 149 - (44/0.8)$

$PAO_2 = 149 - (55)$

$PAO_2 = 94$ mmHg

Now, we can insert our calculated PAO_2 into the A-a Gradient formula.

A-a Gradient = $PAO_2 - PaO_2$

A-a Gradient = 94 - 91

A-a Gradient = 3 mmHg

What is the A-a gradient who has a PaO2 of 84 mmHg and a PaCO2 of 74 mmHg and the PIO_2 is 301 mmHg? (R = 0.8) Express your answer in the nearest whole number.

Examining the problem, we see that the PAO2 isn't provided, so we must calculate it.

$PAO_2 = PIO_2 - (PACO_2/R)$

$PAO_2 = 301 - (74/0.8)$

$PAO_2 = 301 - (92.5)$

$PAO_2 = 208.5$ mmHg

Now, we can insert our calculated PAO_2 into the A-a Gradient formula.

A-a Gradient = $PAO_2 - PaO_2$

A-a Gradient = 208.5 - 84

A-a Gradient = 124.5 mmHg or 125 mmHg

I had mentioned that there is potentially three formulas to solve to calculate the A-a gradient (there's even four if you are required to determine the normal A-a gradient for a patient's age!). In the questions above, we were provided with PIO_2 in mmHg. However, it is unlikely that you have ever decided to increase a patient's inspired oxygen from 150 mmHg to 350 mmHg. You may, however, have decided to increase a patient's FiO2 from 21% to 50%, which is roughly the same thing. (If this sounds confusing, don't worry. I'll explain in just a bit.)

So, where do the values in mmHg come from? Well, that's Step Three. It involves water vapor, barometric pressure, and oxygen concentration. Fortunately, however, the only value we really have to worry about is the FiO2, because the other two stay the same for most of our calculations. (although I may throw a curve in there later just to see if you are paying attention!).

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STEP THREE

We have seen that the PIO_2 is provided in mmHg. If we normally speak of oxygen in terms of percentages, however, then why is it expressed in mmHg, which is a unit of pressure?

It is not just a unit of pressure in this context—it is a unit of *partial* pressure. We are determining the partial pressure of oxygen in relation to the other atmospheric gases like nitrogen. A fact to commit to memory is that normal atmospheric pressure at sea level is 760 mmHg. Another fact that you can commit to memory if you want, is that if you are at sea level, the partial pressure of oxygen in the alveoli is 150 mmHg if breathing 21% oxygen.

So, if the concentration of air is 21%, then we should be able to multiply 21% X 760 mmHg and get the partial pressure of oxygen in the alveoli, right?

PIO_2 = FiO_2 X atmospheric pressure in mmHg

PIO_2 = 0.21 X 760 mmHg

PIO_2 = 159.6 mmHg or 160 mmHg

Wait...I just told you that you that the partial pressure of oxygen in the alveoli (the PIO2) at sea level (760 mmHg) and on room air (21% oxygen) was 150 mmHg and the answer we got in the above calculation was 160 mmHg! Why is this?

As you recall, one of the functions of the respiratory tract is to warm and humidify the air we inhale. Humidifying the air means that it introduces water vapor into the air that we breathe. Now, instead of just breathing oxygen and nitrogen, we are breathing a combination of nitrogen, oxygen, and *water*. Because it is now in a gas form, the water in the alveolar air has it's own partial pressure, which just so happens to be 47 mmHg when the air is completely saturated. So, we must correct our formula to adjust for the presence of water.

PIO_2 = FIO_2 X (atmospheric pressure – water vapor pressure in mmHg)

PIO_2 = 0.21 X (760 – 47)

PIO_2 = 0.21 X (713)

PIO_2 = 149.73 mmHg or 150 mmHg

As you can see, calculating the PIO_2 is not really that complicated as long as you remember to subtract the partial pressure of water vapor from the atmospheric pressure and then multiply that by the FIO_2. So, let's try a few practice examples:

What is the PIO_2 at sea level if the inspired oxygen concentration is 40%?

PIO_2 = FIO_2 X (atmospheric pressure – water vapor pressure in mmHg)

PIO_2 = 0.40 X (760 – 47)

PIO_2 = 0.40 X (713)

PIO_2 = 285.2 mmHg or 285 mmHg

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STEP THREE CONT.

What is the PIO_2 at sea level if the inspired oxygen concentration is 77%?

$PIO_2 = FIO_2 \times (\text{atmospheric pressure} - \text{water vapor pressure in mmHg})$

$PIO_2 = 0.77 \times (760 - 47)$

$PIO_2 = 0.77 \times (713)$

$PIO_2 = 549 \text{ mmHg}$

What is the PIO_2 at sea level if the inspired oxygen concentration is 33%?

$PIO_2 = FIO_2 \times (\text{atmospheric pressure} - \text{water vapor pressure in mmHg})$

$PIO_2 = 0.33 \times (760 - 47)$

$PIO_2 = 0.33 \times (713)$

$PIO_2 = 235.29 \text{ mmHg or } 235 \text{ mmHg}$

What is the PIO_2 at sea level if the inspired oxygen concentration is 55%?

$PIO_2 = FIO_2 \times (\text{atmospheric pressure} - \text{water vapor pressure in mmHg})$

$PIO_2 = 0.55 \times (760 - 47)$

$PIO_2 = 0.55 \times (713)$

$PIO_2 = 392.15 \text{ mmHg or } 392 \text{ mmHg}$

What is the PIO_2 at sea level if the inspired oxygen concentration is 100%?

$PIO_2 = FIO_2 \times (\text{atmospheric pressure} - \text{water vapor pressure in mmHg})$

$PIO_2 = 1.0 \times (760 - 47)$

$PIO_2 = 1.0 \times (713)$

$PIO_2 = 713 \text{ mmHg}$

In the next section, we will perform a few examples that combine the three steps. We will have to begin by calculating the PIO_2, then use that value to calculate the PAO_2, and then we will be able to finally calculate the A-a gradient.

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What is the A-a gradient of a patient who has a FiO_2 of 35%, a PaCO2 of 58 mmHg, and an arterial PO_2 of 95 mmHg? (the patient is at sea level and the R value = 0.8)

The formula for the A-a gradient as you remember, is:

$$A\text{-}a\ Gradient = PAO_2 - PaO_2$$

The arterial partial pressure of oxygen (PaO_2) is provided, but we are not provided with a value for the PAO_2. The next step is to write down the formula to calculate the PAO_2 and see if we have enough information to calculate it:

$$PAO_2 = PIO_2 - (PACO_2/R)$$

We are provided with the value for R in the question (0.8). We don't have a value for $PACO_2$, but we remember that we can substitute the $PaCO_2$ value for the $PACO_2$. Looking back at the question, however, it is evident that the PIO_2 is not provided, so we must calculate it as follows:

$$PIO_2 = FIO_2 \times (atmospheric\ pressure - water\ vapor\ pressure\ in\ mmHg)$$

$$PIO_2 = 0.35 \times (760 - 47)$$

$$PIO_2 = 0.35 \times (713)$$

$$PIO_2 = 249.55\ mmHg$$

Now, we plug the value for PIO_2 into the PAO_2 formula. It is important to note that, although we have been rounding up our answers on each step in the examples, when we work a problem with multiple steps like this one, we do not round up until we reach our *final answer*. This is very important, because in complex problems on your board exam, rounding up at each step introduces a small amount of error that magnifies with each step and could be enough to make the answer you enter on a calculation question incorrect.

$$PAO_2 = PIO_2 - (PACO_2/R)$$

$$PAO_2 = 249.55 - (58/0.8)$$

$$PAO_2 = 249.55 - (72.5)$$

$$PAO_2 = 177.05\ mmHg$$

Finally, we have the information we need to calculate the A-a gradient:

$$A\text{-}a\ Gradient = 177.05\ mmHg - 95\ mmHg$$

$$A\text{-}a\ Gradient = 82.05\ mmHg \text{ or } \mathbf{82\ mmHg}$$

Admittedly, this looks like a lot of work to arrive at a single answer, however, A-a gradients are one of the more complex and time-consuming problems you will face due to the number of steps required to compensate for limited information in the problem. The key is to practice them enough that you remember the formulas and feel confident in your ability to perform them under the stress of an exam.

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What is the A-a gradient of a patient who has a FiO_2 of 70%, a PaCO2 of 44 mmHg, and an arterial PO_2 of 159 mmHg? (the patient is at sea level and the R value = 0.8)

$A\text{-}a\ Gradient = PAO_2 - PaO_2$
$PAO_2 = PIO_2 - (PACO_2/R)$

Looking above at the formulas for A-a gradient and PAO_2, it is apparent that we do not have sufficient information to calculate either directly, so we must start by calculating the PIO_2 and work our way backwards.

$PIO_2 = FIO_2 \times (atmospheric\ pressure - water\ vapor\ pressure\ in\ mmHg)$

$PIO_2 = 0.70 \times (760 - 47)$

$PIO_2 = 0.70 \times (713)$

$PIO_2 = 499.1\ mmHg$

Now, we plug the value for PIO_2 into the PAO_2. Again, remember that we don't round up until the final step.

$PAO_2 = PIO_2 - (PACO_2/R)$

$PAO_2 = 499.1 - (44/0.8)$

$PAO_2 = 499.1 - (55)$

$PAO_2 = 444.1\ mmHg$

Finally, we have the information we need to calculate the A-a gradient:

$A\text{-}a\ Gradient = PAO_2 - PaO_2$

$A\text{-}a\ Gradient = 444.1\ mmHg - 159\ mmHg$

$A\text{-}a\ Gradient = 285.1\ mmHg$ or **285 mmHg**

What is the A-a gradient of a patient who has a FiO_2 of 53%, a PaCO2 of 39 mmHg, and an arterial PO_2 of 301 mmHg? (the patient is at sea level and the R value = 0.8)

$PIO_2 = FIO_2 \times (atmospheric\ pressure - water\ vapor\ pressure\ in\ mmHg)$
$PIO_2 = 0.53 \times (760 - 47)$
$PIO_2 = 0.53 \times (713)$
$PIO_2 = 377.89\ mmHg$

Now, we plug the value for PIO_2 into the PAO_2.

$PAO_2 = PIO_2 - (PACO_2/R)$
$PAO_2 = 377.89 - (39/0.8)$
$PAO_2 = 377.89 - (48.75)$
$PAO_2 = 329.14\ mmHg$

Finally, we have the information we need to calculate the A-a gradient:

$A\text{-}a\ Gradient = PAO_2 - PaO_2$
$A\text{-}a\ Gradient = 329.14\ mmHg - 301\ mmHg$
$A\text{-}a\ Gradient = 28.14\ mmHg$ or 28 mmHg

An 84 year-old patient in the intensive care unit in Denver, Colorado exhibits a PaO_2 of 89 and a

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PaCO2 of 48 on 30% oxygen. Is his A-a gradient normal? (the barometric pressure at Denver is 640 mmHg and R = 0.8)

This problem combines the formulas you have already learned with the complexity of a real-life scenario. Again, it is apparent that we do not have enough information to calculate the A-a gradient or PAO_2 directly and must begin with the PIO_2 calculation. In this problem, we are not at sea level, which means that the value for atmospheric pressure is different. We are given this value in the problem, though.

$PIO_2 = FIO_2 \times (\text{atmospheric pressure} - \text{water vapor pressure in mmHg})$

$PIO_2 = 0.30 \times (640 - 47)$

$PIO_2 = 0.30 \times (593)$

$PIO_2 = 177.9 \text{ mmHg}$

Now, we can fill in the value for PIO_2 in the calculation for PAO_2.

$PAO_2 = PIO_2 - (PACO_2/R)$

$PAO_2 = 177.9 - (48/0.8)$

$PAO_2 = 177.9 - (60)$

$PAO_2 = 117.9 \text{ mmHg}$

Now we can calculate the A-a gradient:

$\text{A-a Gradient} = PAO_2 - PaO_2$

$\text{A-a Gradient} = 117.9 \text{ mmHg} - 89 \text{ mmHg}$

$\text{A-a Gradient} = 28.9 \text{ mmHg or } 29 \text{ mmHg}$

We have calculated the A-a gradient, but this is not what the question asked. The question asked whether or not this value was normal for a patient who was 84 years old. Early in this workbook, I indicated that the formula to estimate the normal A-a gradient based on age was:

$\text{Age-Normal A-a Gradient} = \text{Age}/4 + 4$

$\text{Age-Normal A-a Gradient} = 84/4 + 4$

$\text{Age-Normal A-a Gradient} = 21 + 4$

$\text{Age-Normal A-a Gradient} = 25 \text{ mmHg}$

The normal A-a Gradient for this patient is 25 mmHg and the patient's actual value is 29 mmHg. The patient's A-a gradient is not normal, it is elevated.

The next section includes several practice problems for you to perform on your own with a key at the end of the workbook. Try your best to perform the problems without looking back at the formulas if at all possible. Remember, that if presented with this type of problem on your board exam, you will have access to a calculator, pencil, and paper, but you will have to remember the formulas for them to be of any use to you.

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PRACTICE PROBLEMS

1. What is the A-a gradient of a patient with a PAO_2 of 119 mmHg and a PaO_2 of 98 mmHg?

2. What is the A-a gradient of a patient with a PAO_2 of 212 mmHg and a PaO_2 of 184 mmHg?

3. What is the A-a gradient of a patient with a PAO_2 of 151 mmHg and a PaO_2 of 143 mmHg?

4. What is the A-a gradient of a patient with a PAO_2 of 96 mmHg and a PaO_2 of 68 mmHg?

5. What is the A-a gradient of a patient with a PAO_2 of 307 mmHg and a PaO_2 of 276 mmHg?

6. What is the PAO_2 of a patient with a PIO_2 of 199 mmHg and a $PaCO_2$ of 49 if R=0.8?

7. What is the PAO_2 of a patient with a PIO_2 of 87 mmHg and a $PaCO_2$ of 32 if R=0.8?

8. What is the PAO_2 of a patient with a PIO_2 of 208 mmHg and a $PaCO_2$ of 43 if R=0.8?

9. What is the PAO_2 of a patient with a PIO_2 of 113 mmHg and a $PaCO_2$ of 38 if R=0.8?

10. What is the PAO_2 of a patient with a PIO_2 of 184 mmHg and a $PaCO_2$ of 31 if R=0.8?

11. What is the PIO_2 of a patient at sea level if the FiO_2 is 99%?

12. What is the PIO_2 of a patient at sea level if the FiO_2 is 24%?

13. What is the PIO_2 of a patient at sea level if the FiO_2 is 45%?

14. What is the PIO_2 of a patient at sea level if the FiO_2 is 55%?

15. What is the PIO_2 of a patient at sea level if the FiO_2 is 80%

16. What is the A-a gradient of a patient who has a FiO_2 of 55%, a $PaCO_2$ of 51 mmHg, and an arterial PO_2 of 186 mmHg? (the patient is at sea level and the R value = 0.8)

17. What is the A-a gradient of a patient who has a FiO_2 of 40%, a $PaCO_2$ of 62 mmHg, and an arterial PO_2 of 95 mmHg? (the patient is at sea level and the R value = 0.8)

18. What is the A-a gradient of a patient who has a FiO_2 of 24%, a $PaCO_2$ of 43 mmHg, and an arterial PO_2 of 101 mmHg? (the patient is at sea level and the R value = 0.8)

19. What is the A-a gradient of a patient who has a FiO_2 of 75%, a $PaCO_2$ of 31 mmHg, and an arterial PO_2 of 205 mmHg? (the patient is at sea level and the R value = 0.8)

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20. What is the A-a gradient of a patient who has a FiO_2 of 50%, a $PaCO_2$ of 39 mmHg, and an arterial PO_2 of 144 mmHg? (the patient is at sea level and the R value = 0.8)

21. What is the A-a gradient of a patient who has a FiO_2 of 40%, a $PaCO_2$ of 45 mmHg, and an arterial PO_2 of 150 mmHg? (the patient is at sea level and the R value = 0.8)

22. What is the A-a gradient of a patient who has a FiO_2 of 60%, a $PaCO_2$ of 42 mmHg, and an arterial PO_2 of 229 mmHg? (the patient is at sea level and the R value = 0.8)

23. What is the A-a gradient of a patient who has a FiO_2 of 70%, a $PaCO_2$ of 56 mmHg, and an arterial PO_2 of 180 mmHg? (the patient is at sea level and the R value = 0.8)

24. What is the A-a gradient of a patient who has a FiO_2 of 30%, a $PaCO_2$ of 37 mmHg, and an arterial PO_2 of 114 mmHg? (the patient is at sea level and the R value = 0.8)

25. What is the A-a gradient of a patient who has a FiO_2 of 35%, a $PaCO_2$ of 60 mmHg, and an arterial PO_2 of 133 mmHg? (the patient is at sea level and the R value = 0.8)

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ANSWER KEY

1. 21 mmHg
2. 28 mmHg
3. 8 mmHg
4. 28 mmHg
5. 31 mmHg
6. 138 mmHg
7. 47 mmHg
8. 154 mHg
9. 66 mmHg
10. 145 mmHg
11. 706 mmHg
12. 171 mmHg
13. 321 mmHg
14. 392 mmHg
15. 570 mmHg
16. PIO_2 = 392.15 mmHg, PAO_2 = 328.35 mmHg, Aa Gradient = 142 mmHg
17. PIO_2 = 285.2 mmHg, PAO_2 = 207.7 mmHg, Aa Gradient = 112 mmHg
18. PIO_2 = 171.12 mmHg, PAO_2 = 117.37 mmHg, Aa Gradient = 16 mmHg
19. PIO_2 = 534.75 mmHg, PAO_2 = 496 mmHg, Aa Gradient = 291 mmHg
20. PIO_2 = 356.5 mmHg, PAO_2 = 307.75 mmHg, Aa Gradient = 164 mmHg
21. PIO_2 = 285.2 mmHg, PAO_2 = 228.95 mmHg, Aa Gradient = 79 mmHg
22. PIO_2 = 427.8 mmHg, PAO_2 = 375.3 mmHg, Aa Gradient = 146 mmHg
23. PIO_2 = 499.1 mmHg, PAO_2 = 429.1 mmHg, Aa Gradient = 249 mmHg
24. PIO_2 = 213.9 mmHg, PAO_2 = 167.65 mmHg, Aa Gradient = 54 mmHg
25. PIO_2 = 249.55 mmHg, PAO_2 = 174.55 mmHg, Aa Gradient = 42 mmHg

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